

kmhelpers: A Python toolkit for automated management of genomic indexes

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Summary

Large-scale genomic sequence search has become a central problem in modern bioinformatics (Karasikov et al., 2025; Marchet, 2024). A widely used family of methods relies on Bloom filters (BF) (Bloom, 1970) to record, for each indexed sample, which k -mers (words of length k) it contains; a query then reports the set of samples containing each queried k -mer. Building such an index over many samples is not a single operation but a chain of interdependent steps, and each step demands specialist knowledge: sizing each Bloom filter to its sample, configuring the third-party components used internally by `kmindex` (such as the `findere` (Robidou & Peterlongo, 2021) approximate-membership layer), distributing data and computation, bounding peak RAM, and grouping samples into sub-indexes so that the final index stays small without slowing queries. An error at any step can invalidate downstream results, waste significant computation, or produce an oversized index.

We present `kmhelpers`, an open-source Python toolkit that automates this entire workflow for indexes built with `kmindex` (Lemane et al., 2024). It hides the technical decisions listed above behind a command-line interface (CLI) and a matching Python API that cover every stage of the k -mer index lifecycle, from raw samples up to queries, including the federation of independently built indexes into a single queryable resource.

Statement of Need

k -mer-based sequence databases built with tools such as `kmindex` (Lemane et al., 2024) and `kmtricks` (Lemane et al., 2022) answer queries on large-scale samples in genomics, metagenomics, and population studies. They were recently used to index 50 petabytes of raw sequencing data (Chikhi et al., 2024). Yet a wide gap separates having the indexing software installed from running a reproducible, end-to-end, size-optimized indexing workflow.

Crossing that gap currently requires a user to master, simultaneously: Bloom filter sizing as a function of each sample's k -mer cardinality (number of k -mers to be indexed per sample); configuring internal components of `kmindex`; distributing data and computations; controlling the peak memory; and partitioning the samples into sub-indexes that balance index size against query speed. This expertise is not required to obtain a functional index but to obtain an *efficient* one that minimizes index size and maximizes query speed under a fixed false-positive-rate constraint.

`kmhelpers` removes that barrier. It makes the complete process of building, updating, and querying a `kmindex` search engine accessible to any group (research teams, sequencing platforms, hospitals, or data-production centers) that holds a potentially large genomic dataset, whether the resulting index is kept private or shared, without repeated manual re-optimization as the dataset grows. This opens to non-specialists both the creation and maintenance of optimized

41 indexes.

42 Indexes are rarely static: as new samples accumulate, they must be regularly updated.
43 kmhelpers therefore also offers an update feature for easily adding new samples to an existing
44 index.

45 Formally, the input is a set $\mathcal{S} = \{S_1, \dots, S_n\}$ of n genomic samples of various sizes; each S_i
46 is either a raw sequencing dataset or assembled sequences. The output is a kmindex index
47 subject to two user-defined limits: (1) the maximum allowed false-positive rate, and (2) the
48 maximum number of sub-indexes, each a file storing a set of BFs as a matrix. kmhelpers
49 orchestrates every step between input and output, automating the parametrization and the
50 data distribution.

51 State of the Field

52 Several tools tackle large-scale sequence search with k -mer-based data structures. BIGSI
53 (Bradley et al., 2019) and COBS (Bingmann et al., 2019) use compressed Bloom-filter matrices
54 to answer presence/absence queries across collections of sequencing experiments. HowDeSBT
55 (Harris & Medvedev, 2020) and Mantis (Pandey et al., 2018) rely on sequence Bloom trees and
56 counting quotient filters, respectively. More recent colored- k -mer indexes such as MetaGraph
57 (Karasikov et al., 2025) and Fulgor (Fan et al., 2024) push scalability further; see (Marchet,
58 2024) for a survey. kmindex, which kmhelpers wraps, builds on the kmtricks (Lemane et al.,
59 2022) counting pipeline and is designed for efficient querying of large sequence collections.

60 kmhelpers does not compete with any of these approaches at the algorithmic level. Its
61 contribution lies in correctly parametrizing kmindex's individual components and orchestrating
62 them into a single, size-optimized workflow. kmhelpers implements the decisions a kmindex
63 expert would otherwise make by hand: profile computes Bloom filter sizes and sample-to-
64 sub-index assignments that minimize total index size under a false-positive-rate constraint;
65 plan estimates disk and memory requirements before any build starts; and registry federates
66 independently built sub-indexes into one logical index. No dedicated automation layer of this
67 kind existed for kmindex prior to kmhelpers.

68 Software design

69 Background: the sub-index structure

70 A Bloom-filter index stores all BFs of the same size as a single (row-major) matrix in one file.
71 In such a matrix each column is one BF (one sample) and each row records the presence (1)
72 or absence (0) of a k -mer across all samples sharing that filter size, assuming a common hash
73 function. A complete index is therefore a set of such matrices, each one a *sub-index* holding
74 all BFs of a given size.

75 The difficulty is that each BF size must match the number of items it stores, here the distinct
76 k -mers of a sample. If all samples had the same number of distinct k -mers, every BF would
77 share one size and a single matrix would suffice. This is the ideal case, where only one file is
78 opened per query and one matrix row gives the answer across all n samples. In practice, sample
79 sizes differ by several orders of magnitude. Sizing every BF for the largest sample wastes
80 enormous storage. Giving each sample its own single-column matrix minimizes storage but
81 forces a query to open n files (potentially millions), severely slowing down queries. kmhelpers
82 takes the middle ground: the user fixes the maximum number of sub-indexes, and the profile
83 step chooses the per-sub-index BF size that minimizes total index size under that constraint,
84 before distributing each input sample to its respective sub-index.

The pipeline

kmhelpers exposes the index lifecycle as a sequence of commands, illustrated Figure 1:

- **list** — recursively discovers all samples in a given directory and counts each sample's distinct k -mers using ntCard (Mohamadi et al., 2017) (unless the counts are provided by the user).
- **profile** — determines the best set of sub-index BF sizes given the user-defined maximum number of sub-indexes and target false-positive rate.
- **compose** — assigns each sample to its sub-index and generates the *files-of-files* describing the data origin of each sub-index.
- **plan** — validates sample files, available disk space, and memory upfront, and emits ready-to-execute pipeline scripts.
- **apply** — builds all sub-indexes by invoking kmindex, with span-level and name-level filtering.

For ease of use, steps **list**, **profile**, and **compose** can be grouped under a single command named **design**, and steps **plan** and **apply** can be grouped under the **build** command.

Once an index is built, kmhelpers also answers queries (**query**). Multi-step workflows can be described as declarative YAML pipelines (**pipeline**) and executed in a single command.

An additional command, **registry**, lets users register several distinct indexes (built locally or hosted anywhere accessible) into one logical index, redirecting each query to all registered indexes at query time.

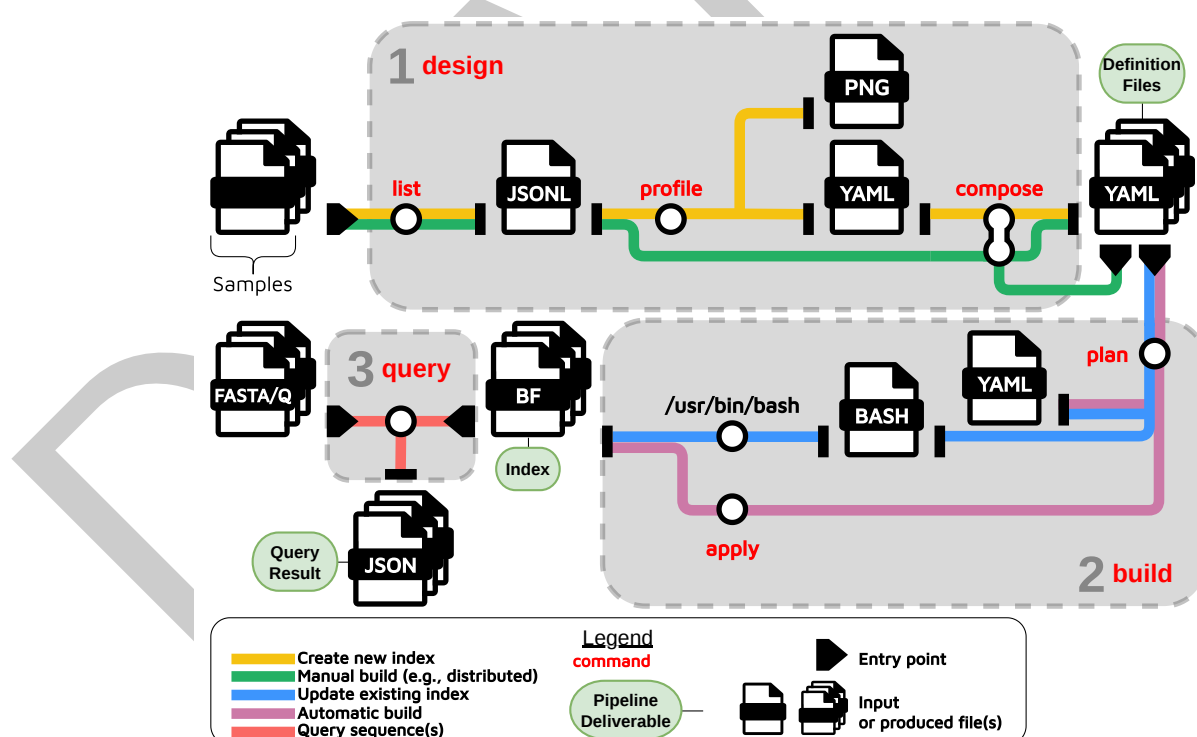


Figure 1: Overview of the kmhelpers workflow

Implementation

kmhelpers is implemented in Python (≥ 3.8) and distributed via Conda, which installs its bioinformatics dependencies (kmindex, ntCard) automatically. The CLI is built with Click (Ronacher & Click Contributors, 2023), and the package exposes a public Python API

109 covering all CLI functionality. Together, the declarative YAML index-definition format and
110 the plan/apply separation make k -mer indexing workflows accessible to researchers who are
111 not experts in the underlying data structures, while remaining flexible enough for large-scale
112 production use.

113 Research impact statement

114 Applying kmhelpers to the “Tree of Life” dataset

115 In phase two, the Earth Biogenome Project aims to sequence reference genomes of 150,000
116 family representatives across the tree of life over the next four years (Blaxter et al., 2025). Tree
117 of Life Programme at the Wellcome Sanger Institute is one of the major contributors to this
118 effort with large sequencing projects, such as Darwin Tree of Life for all species found in Britain
119 and Ireland (Darwin Tree of Life Project Consortium, 2022). To date, Tree of Life has already
120 released more than 4,000 genomes and is in the process of sequencing an additional 3,000.
121 The project is based on robustly identified specimens, but occasionally long-read datasets and
122 chromatin capture (HiC) sequencing do not match, which might happen due to cryptic species,
123 sequencing hybrid specimens, or mistakes in sample handling either by the collector or during
124 the sequencing process. These cases we refer to as sample swaps.

125 This project is a perfect use case for applying kmhelpers: from input data — here a mixture of
126 raw sequencing reads and assembled genomes — to a final index powering a CLI search engine
127 that is now being used for identification of sample swaps (matching sequencing libraries) at
128 scale. Concretely, kmhelpers was applied to 7,394 samples and this project also helped shape
129 the design of kmhelpers itself. The index is continuously updated as new sequencing data are
130 progressively added, which is essential for sustaining detection of sample swaps.

131 Using kmhelpers for updating the Logan-Search search engine

132 The Logan-Search search engine (Chikhi et al., 2024) indexes approximately 50 petabytes of
133 sequence data from SRA (Katz et al., 2022). Since its creation, the SRA database size has
134 doubled. The index of Logan-Search will be updated using kmhelpers, also optimizing the
135 sub-indexes design.

136 Availability and documentation

137 kmhelpers is released under the GNU General Public License v3.0 and is available at <https://github.com/seblins/kmhelpers>. The full documentation is available at <https://seblins.github.io/kmhelpers/>.

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147 AI Usage Disclosure

148 AI assistance was used for constrained tasks (code suggestions and drafting the paper) under
149 strict human review at every stage. AI provider used: Claude (Anthropic, 2025).

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